



Fig. 7: An anti-satellite weapon at work on its target, a representative picture (Source: <https://www.amarujala.com>)

development programme, an exo-atmospheric interceptor, Prithvi Defence Vehicle (PDV), has been developed. In 2017, the first real-time interception test Mission Shakti was carried out against a manoeuvring target.

India announced during the 97th Indian Science Congress held in 2009 at the University of Kerala that it was working on operation Mission Shakti to develop the technology to produce a weapon to destroy enemy satellites in orbit. On March 27, 2019 India conducted an anti-satellite missile test. The test required an extremely high degree of precision and technical capability.

The satellite used in the mission as a target was one of India's existing satellites operating in low orbit. The ASAT weapon was launched at 05:40 UTC (Coordinated Universal Time. IST = UTC + 05.30) at the Integrated Test Range (ITR) in Abdul Kalam Island. It hit and destroyed the test satellite at a 283km altitude in low earth orbit after 168 seconds. The test made India the fourth country after the United States, Russia, and China to have tested an ASAT weapon. The test was fully successful and achieved all parameters as per plan.

The interceptor used in the test was a modified PDV Mark-II missile, which had a hit-to-kill capable Kinetic Kill vehicle. The missile, which was a direct-ascent anti-satellite weapon, was 13 metres long with a diameter of 1.4 metres. It was capa-

ble of shooting down targets moving at a speed of 10kmps at an altitude of 1200km. Being a three-stage missile, it was fitted with two solid-propellant rocket motor stages and the Kill Vehicle. The combined weight of the first and the second stages was 17.2 tons, with the third stage weighing 1.8 tons.

The first two stages, which carried 16.7 tons of fuel, took the missile to the required height and velocity. After that, the nose tip heat shield was ejected and the IIR (imaging infrared) seeker of the Kill Vehicle locked onto the target satellite. The target was hit with an accuracy of within 10cm, which is comparable with the best reported performance of ASAT missiles.

The Kill Vehicle had an advanced terminal guidance system on board, which included a non-gimballed strap-down IIR seeker and an inertial navigation system that used ring-laser gyroscopes. Course corrections to the Kill Vehicle's trajectory were made using a thrust-vector system. The Kill Vehicle could de-

stroy targets in the entire low Earth orbit region.

DRDO said that it is possible to increase the propulsive power of the interceptor to make it capable of targeting satellites at medium altitudes. To minimise the threat of debris, the interception was performed at an altitude below 300km.

Space debris

The test was conducted at low altitude to ensure that the resulting debris would decay and fall back onto the earth atmosphere within weeks. The impact generated more than 400 pieces of orbital debris. According to an initial assessment of DRDO, some of the debris, depending on size and trajectory, were estimated to re-enter in 45 days.

NASA's assessment was that the debris could last for years because the solar minimum had contracted the atmosphere that would otherwise cause the debris to re-enter. Jonathan McDowell, an astrophysicist at the Harvard-Smithsonian Centre for Astrophysics, was of the view that most debris should burn up in the atmosphere within weeks, while some might persist for a year. Brian Weeden of Secure World Foundation, however, warned about the possibility of some fragments getting boosted to higher orbits.

Indian missile experts said that the debris present in a low orbit do not pose a threat to any spacecraft. As of September 26, 2019, there were fifty tracked pieces of debris in orbit, nine of which were lost according to astrodynamics expert T.S. Kelso.

Opportunities opening for the industry

The defence sector has been identified as one of the core areas to boost *Atmanirbhar Bharat* (self-reliant India) programme, with the vision of a turnover of \$25 billion by 2025. Under *Atmanirbhar Bharat* a positive indigenisation list of 209 items has been prepared with an embargo on



Fig. 8: An exo-atmospheric Kill Vehicle (Source: <https://defense-update.com>)